Infosys Array & String Preparation Guide (Detailed)

1. Largest Element in Array

Input: 5 2 9 1 7
Output: 9

```
int max=arr[0];
for(int i=1;i<arr.length;i++)
 if(arr[i]>max) max=arr[i];
```

2. Second Largest Element

Input: 5 2 9 1 7
Output: 7

```
int max=Integer.MIN_VALUE, second=Integer.MIN_VALUE;
```

3. Reverse Array

Input: 1 2 3 4
Output: 4 3 2 1

```
for(int i=arr.length-1;i>=0;i--) System.out.print(arr[i]);
```

4. Array Rotation

Input: 1 2 3 4 5
Output: 3 4 5 1 2

```
Left rotate by k positions using loops.
```

5. Missing Number

Input: 1 2 3 5
Output: 4

```
Use sum formula n*(n+1)/2.
```

6. Palindrome String

Input: madam
Output: Palindrome

```
Compare string with its reverse.
```

7. Reverse String

Input: Java
Output: avaJ

```
Traverse from end to start.
```

8. Anagram

Input: listen silent
Output: Anagram

```
Sort both strings and compare.
```

9. String Rotation

Input: ABCD CDAB
Output: True

```
(s1+s1).contains(s2)
```

10. Remove Duplicates

Input: programming
Output: progamin

```
Use temp string and indexOf().
```

11. Character Frequency

Input: banana

```
Count occurrences using nested loops.
```

12. Count Vowels

Input: education
Output: 5

```
Check against a,e,i,o,u.
```

13. First Non-Repeating Character

Input: swiss
Output: w

```
Frequency count then first freq=1.
```

14. Longest Substring Without Repeating Characters

Input: JavaProgramming

```
Use temp string and indexOf().
```

15. Triple String Transformation

Input: apple bread orange
Output: %ppl%##ea#ORANGE

```
Replace vowels, consonants, uppercase.
```

16. Email Validation

Input: abc@gmail.com

```
Regex: [a-zA-Z0-9._]+@[a-zA-Z]+\.[a-zA-Z]{2,}
```

17. Move Zeros to End

Input: 1 0 2 0 3
Output: 1 2 3 0 0

```
Shift non-zero elements first.
```

18. Pair With Given Sum

Input: [1,2,3,4], sum=5

```
Check all pairs using loops.
```

19. Maximum Occurring Character

Input: banana
Output: a

```
Track highest frequency.
```

20. String Compression

Input: aaabbc
Output: a3b2c1

```
Count consecutive characters.
```

Infosys Interview Tips

1. Practice arrays and strings daily.
2. Learn nested loops thoroughly.
3. Avoid excessive built-in methods.
4. Understand time complexity basics.
5. Be able to explain your logic clearly.
6. Practice dry runs on paper.

Practice Sheet 1

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 2

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 3

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 4

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 5

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 6

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 7

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 8

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 9

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.

Practice Sheet 10

Solve: Reverse String, Palindrome, Anagram, Rotation, Frequency Count, Largest Element, Second Largest, Missing Number, Array Rotation, Triple String Transformation.